MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, K11 OPERATION AND SEQUENCE

1. SCOPE

1.1 This specification covers the detail requirements of one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-60898 comprise the complete requirements for the K11 Operation and Sequence Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-60898	Circuit Card Assemblies, General Specification for

DRAWINGS

U. S. Army, Frankford Arsenal

F10554165	Electronic Component Assembly, K11 Operation and Sequence

PACKAGING DATA SHEET

10554165	Packaging of Circuit Card Assembly, K11 Operation and Sequence

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 4931

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3. REQUIREMENTS

3.1 Fabrication. The assembly shall be manufactured in accordance with Drawing F10554165 and drawings pertaining thereto.

3.2 General specification. The assembly shall meet the requirements of MIL-C-60898.

3.3 Environmental service conditions. See MIL-C-60898.

3.3.1 Performance at temperature extremes. For the conditions of 3,3.2 of MIL-C-60898, the requirements of 3,4 shall be met with exceptions as specified in the following subparagraphs.

3.3.1.1 Logic "1" output. A logic "l" output shall be greater than +2.5 vdc but less than +4.4 vdc.

3.3.1.2 Logic "O" output. A logic "O" output shall be greater than +0.01 vdc but less than +0.4 vdc.

3.3.1.3 Pin 27 output. The output voltage of pin 27 shall be either +20.5 ± 1.5 vdc or a level greater than +0.05 vdc but less than +1.2 vdc.

3.4 Electrical

3.4.1 Conditions of operation. The requirements of 3.4.2 and 3.4.3 shall be met with the input terminations to the board as shown in Figure 1. A logic "0" input shall be +0.50 ± 0.01 vdc and a logic "1" input shall be +2.80 ± 0.01 vdc.

3.4.2 Logic outputs

3.4.2.1 output pin 15. With the inputs applied individually, as specified in Table 1, the output between pin 15 (+) and pin 1 (-) shall be either a logic "O" or a logic "l", as specified in Table 1, A logic "1" output shall be greater than +2.8 vdc but less than +4.4 vdc. A logic "O" output shall be greater than +0.01 vdc but less than +0.4 vdc.

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Table I Output Logic, Pin 15

Input pins			Output pin
5	7	9	15
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0

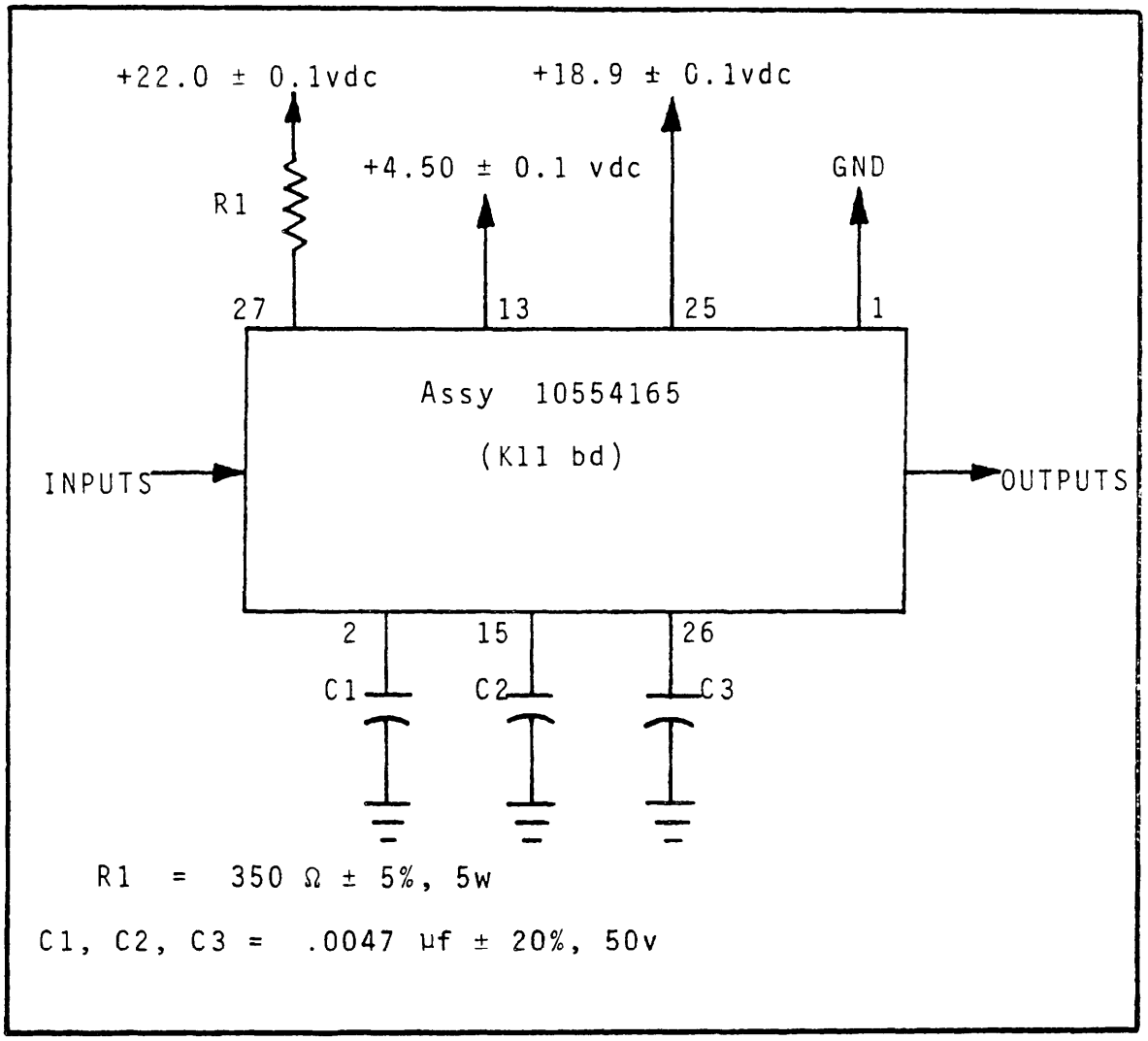


Figure 1

Circuit Card K11 Terminations

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3.4.2.2 Output pin 18. With the inputs applied individually as specified in Table II, the output between pin 18 (+) and pin 1 (-) shall be either a logic "0" or a logic "1" as specified in Table II.

Table II Output Logic. Pin 18

Input pins			Output pin
19	21	23	18
1	1	0	1
0	1	0	0
0	0	1	1
1	1	1	1

3.4.3 Sequence and timing. With the inputs applied sequentially as specified in Table III, the outputs at pin 2 and pin 27 shall be as specified in Table III. Pin 1 is logic ground.

Table III Sequence and Timing logic

Step	Input Pins					Output Pins	
	3	5	7	9	23	2	27
1	0	**	1	1	1	0	+21.5 ± 0.5 vdc
2	0	1	0	1	1	0	+21.5 ± 0.5 vdc
3	0	1	1	1	1	1*	+21.5 ± 0.5 vdc
4	1	1	1	1	1	1	+21.5 ± 0.5 vdc
5	0	1	1	1	1	1	+0.05 to +1.2 vdc
6	0	**	1	1	1	0	+21.5 ± 0.5 vdc

* Logic "1" delayed 1000 ± 50 milliseconds from time pin 7 goes to a logic "1".

** Apply a logic "0" for one to five seconds and then apply a logic "1".

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3.4.4 Capacitor C2. With the signal shown in Figure 2 applied through a 100 ohm ± 10 percent resistor, between pin 13 (+) and pin 1, the voltage out between pin 13 (+) and pin 1 shall be as shown in Figure 3.

3.4.5 Capacitor C1. With the signal shown in Figure 2 applied through a 100 ohm ± 10 percent resistor, between pin 25 (+) and pin 1, the voltage out between pin 25 (+) and pin 1 shall be as shown in Figure 3.

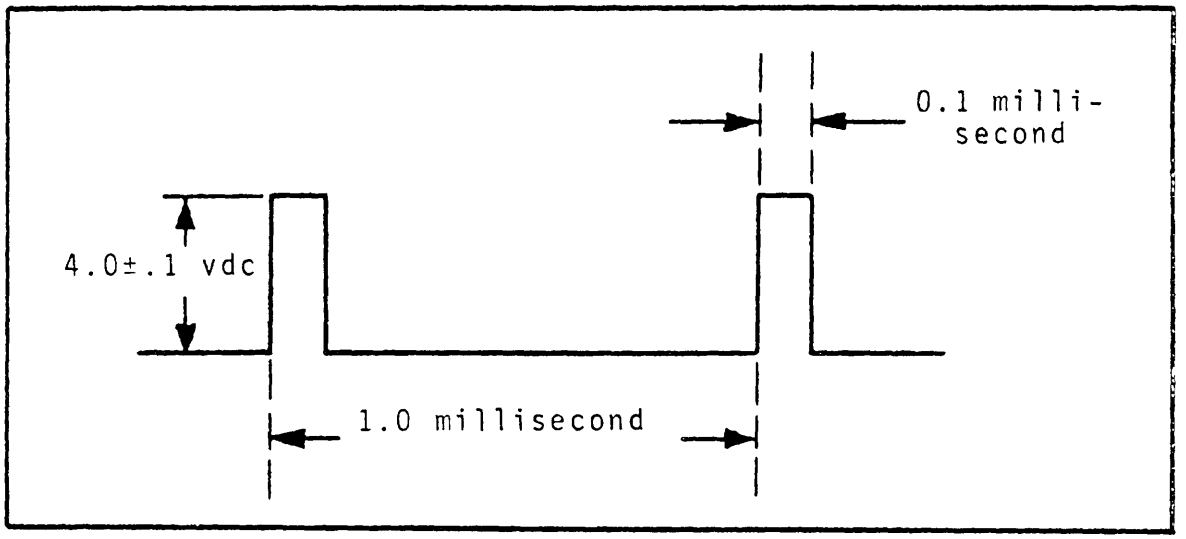


Figure 2

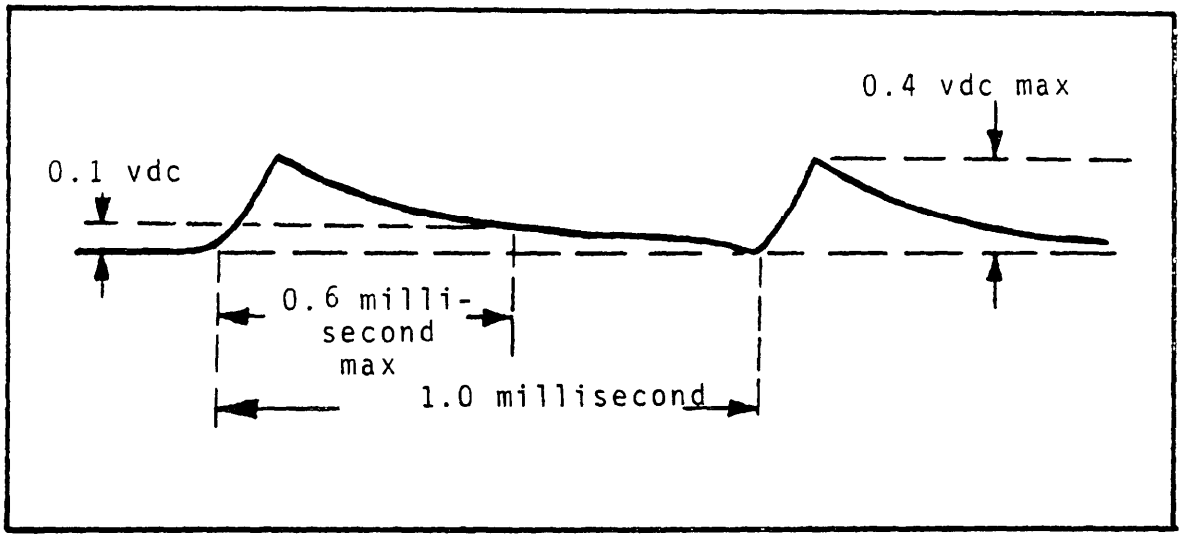


Figure 3

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4. QUALITY ASSURANCE PROVISIONS

4.1 General. The quality assurance provisions of MIL-C-60898 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. As specified in MIL-C-60898.

4.3 Control test sample. As specified in MIL-C-60898.

4.4 Functional tests. The following characteristics shall be tested 100 percent in accordance with the applicable test method.

	Characteristic	Requirement	Procedure
	Logic outputs	3.4.2	4.5.2.2
100	Output pin 15	3,4.2.1	4.5.2.2.1
101	Output pin 18	3.4.2.2	4,5.2.2.2
102	Sequence and timing	3.4.3	4.5.2.3
103	Capacitor C2	3.4.4	4.5.2.4
104	Capacitor C1	3.4.5	4.5.2.5

4.4.1 Inspection equipment. As specified in MIL-C-60898 and 4.4.1.1.

4.4.1.1 Test equipment. In conjunction with 4.4.1 the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5

	Item	Description
1.	POWER SUPPLY	+22.0 ± 0.1 vdc, no less than 500 ma., no greater than 100 mv rms ripple.
2.	POWER SUPPLY	+4.5 ± 0.1 vdc, no less than 500 ma., no greater than 100 mv rms ripple.
3.	POWER SUPPLY	+18.9 ± 0.1 vdc, no less than 500 ma , no greater than 100 mv rms ripple.
4.	POWER SUPPLY	+0.50 ± 0.01 vdc no less than 50 ma , no greater than 5 mv rms ripple.
5.	POWER SUPPLY	+2.80 ± 0.01 vdc, no less than 50 ma, no greater than 5 mv rms ripple.

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.	DIGITAL VOLTMETER	Accuracy and resolution not less than 0.2 percent from 0.0090 to 99.0 vdc. Input impedance no less than 10 megohms on all voltage scales.
'.	OSCILLOSCOPE	Voltage, time accuracy and resolution no less than 5 percent. Rise time no greater than 100 nano seconds. Sweep periods from one microsecond per centimeter to one second per centimeter. Amplitude scales from 0.01 to 10.0 volts per centimeter. Automatic triggering.
3.	ASSORTED COMPONENTS	Resistor (1) 350 ± 5%, 5 watt (R_1). Capacitor (3) 0.0047 ± 20% uf, 50 vdc (Cl, C2 and C3). One resistor, 100 ohms ± 10 percent, ½ watt.
9.	PULSE GENERATOR	Frequency - 1 kilohertz minimum. Duty factor - 10 percent minimum. Amplitude - a minimum of 4 volts into a 200 ohm load. Pulse rise and fall times - less than 0.2 microseconds.

4.5 Test methods and procedures

4.5.1 Environmental. As specified in MIL-C-60898 except the requirements of 3.3.1 herein shall be verified during temperature extremes.

4.5.2 Electrical tests

4.5.2.1 Conditions of operation. Unless otherwise specified, all subsequent tests are to be performed with the conditions of operation specified in 3.4.1. Connect Rl, Cl, C2, and C3, of the ASSORTED COMPONENTS, item 8 of 4.4.1.1, as shown in Figure 1. Connect the POWER SUPPLIES, items 1, 2, and 3 of 4,4.1.1, with the negative terminals connected to GND as shown in Figure 1. When required during subsequent tests, use the positive output of POWER SUPPLY, item 4 of 4.4.1.1, for logic "0" and the positive output of POWER SUPPLY, item 5, for logic "1". Connect the negative terminals of POWER SUPPLY, item 4 and 5 of 4.4.1.1, to GND as shown in Figure 1.

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4.5.2.2 Logic outputs. In the following tests, the DIGITAL VOLTMETER, item 6 of 4.4.1.1, shall be used to verify that the logic-outputs are within the tolerances specified in 3.4.2.

4.5.2.2.1 Output pin 15. Apply logic "O" and logic "1" inputs, as shown in Table I, to pins 5, 7, and 9. Using the DIGITAL VOLTMETER, item 6 of 4.4.1.1, verify that the output voltage on pin 15 conforms to the requirements of 3.4.2.1 and Table I for each combination of inputs to pins 5, 7, and 9.

4.5.2.2.2 Output pin 18. Apply logic "O" and logic "1" inputs as shown in Table II, to pins 19, 21, and 23. Using the DIGITAL VOLTMETER, item 6 of 4.4.1.1, verify that the output voltage on pin 18 conforms to the requirements of 3.4.2.2 and Table II for each combination of inputs on pins 19, 21, and 23.

4.5.2.3 Sequence and timing. Apply logic "O" and logic "1" inputs, as shown in Table III for each step, to pins 3, 5, 7, 9, and 23. Connect the OSCILLOSCOPE, item 7 of 4.4.1.1, between pins 2 (+) and 1 (-). Connect the OSCILLOSCOPE trigger input between pin 7 (+) and pin 1(-). Adjust the OSCILLOSCOPE to trigger when pin 7 changes from logic "O" to logic "1". Using the DIGITAL VOLTMETER, verify that the output voltage on pins 2 and 27 conform to the requirements of 3.4.3 and Table III for each step. Verify, on the OSCILLOSCOPE that pin 2 changes to logic "1" within 1000 ± 50 milliseconds after pin 7 changes to logic "1" as specified in step 3 of Table 111.

4.5.2.4 Capacitor C2. Connect the PULSE GENERATOR, item 9 of 4.4.1.1, thru the 100 ohm RESISTOR, item 8 of 4.4.1.1, and connect channel A verticle input of OSCILLOSCOPE, item 7 of 4.4.1.1, between pin 13 (+) and pin 1 per Figure 4. Adjust the PULSE GENERATOR to produce the pulse of Figure 2. Connect the OSCILLOSCOPE channel B verticle input, per Figure 4, and verify the waveform of Figure 3.

4.5.2.5 Capacitor Cl, Connect the PULSE GENERATOR, item 9 of 4.4.1.1, thru the 100 ohm RESISTOR, item 8 of 4.4.1.1, and connect channel A verticle input of OSCILLOSCOPE, item 7 of 4.4.1.1, between pin 25 (+) and pin 1 per Figure 5. Adjust the PULSE GENERATOR to produce the pulse of Figure 2. Connect the OSCILLOSCOPE channel B verticle input, per Figure 5, and verify the waveform of Figure 3.

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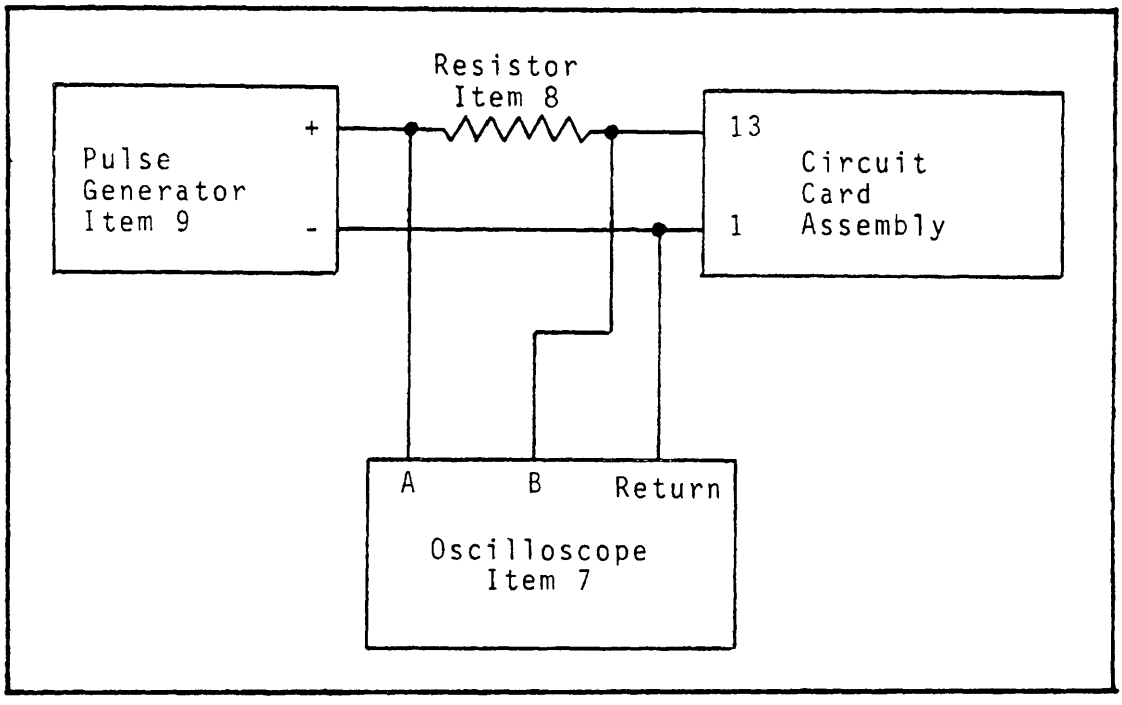


Figure 4

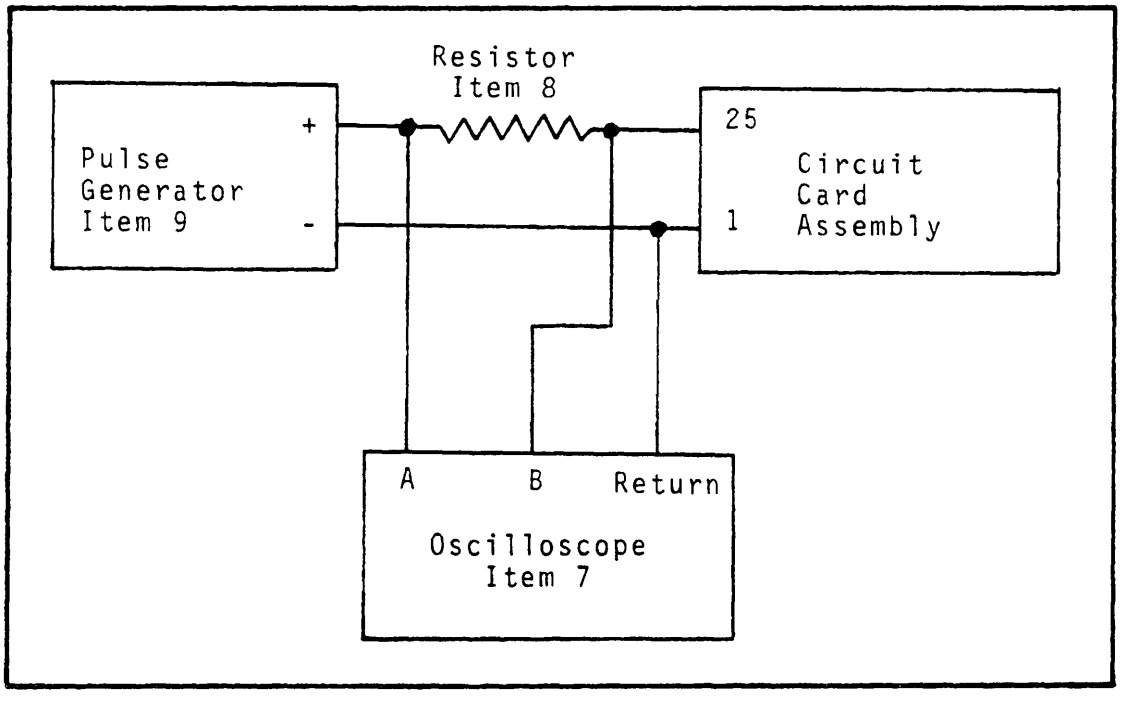


Figure 5

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with Packaging Data Sheet 10554165.

6, NOTES

6.1 The notes specified in MIL-C-60898 are applicable to this specification.

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

( ) DIRECT GOVERNMENT CONTRACT ( ) SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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